

PESU EVENT AND RESOURCE MANAGEMENT

Mini Project – UE23CS351A

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1. TRIGGER

Question: Automatically update resource availability when booked.

Code:

DELIMITER \$\$

```
CREATE TRIGGER trg_resource_booking_check
BEFORE INSERT ON tbl_event_resources
FOR EACH ROW
BEGIN
    DECLARE available_qty INT;
    DECLARE available_status TINYINT;
    DECLARE msg TEXT DEFAULT NULL;
    SELECT quantity, is_available INTO available_qty, available_status
    FROM tbl_resources
    WHERE id = NEW.resource_id;
    IF available_qty IS NULL THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: Resource not found.';
    END IF;
    IF available_status = 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: Resource is currently unavailable for
booking.';
    END IF;
    IF NEW.quantity_booked > available_qty THEN
        SET msg = CONCAT('Error: Only ', available_qty, ' units available
for this resource.');
```

```
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = msg;
    END IF;
    UPDATE tbl_resources
    SET
        quantity = quantity - NEW.quantity_booked,
        is_available = CASE WHEN (quantity - NEW.quantity_booked) <= 0
    THEN 0 ELSE 1 END,
        maintenance_status = 'Booked'
    WHERE id = NEW.resource_id;
END$$

DELIMITER ;
```

Output Screenshot:

```
mysql> select * from tbl_resources;
```

id	name	type	quantity	description	is_available	maintenance_status
1	Projector Screen	AV Equipment	10	Portable 100-inch projector screens	1	Available
2	Microphone Set	AV Equipment	20	Wireless microphone with stand	1	Available
3	Tables	Furniture	100	6-foot rectangular tables	1	Available
4	Chairs	Furniture	500	Standard plastic chairs	0	Under Maintenance
5	Sound System	AV Equipment	5	High-power speakers with mixer	1	Available

5 rows in set (0.00 sec)

```
mysql> INSERT INTO tbl_event_resources (event_id, resource_id, quantity_booked, booking_start, booking_end)
-> VALUES (1, 4, 200, '2025-10-28 09:00:00', '2025-10-28 17:00:00');
ERROR 1644 (45000): Error: Resource is currently unavailable for booking.
mysql> INSERT INTO tbl_event_resources (event_id, resource_id, quantity_booked, booking_start, booking_end)
-> VALUES (1, 5, 1, '2025-10-28 09:00:00', '2025-10-28 17:00:00');
Query OK, 1 row affected (0.04 sec)

mysql> select * from tbl_resources;
```

id	name	type	quantity	description	is_available	maintenance_status
1	Projector Screen	AV Equipment	10	Portable 100-inch projector screens	1	Available
2	Microphone Set	AV Equipment	20	Wireless microphone with stand	1	Available
3	Tables	Furniture	100	6-foot rectangular tables	1	Available
4	Chairs	Furniture	500	Standard plastic chairs	0	Under Maintenance
5	Sound System	AV Equipment	4	High-power speakers with mixer	1	Booked

5 rows in set (0.00 sec)

```
mysql> INSERT INTO tbl_event_resources (event_id, resource_id, quantity_booked, booking_start, booking_end)
-> VALUES (1, 2, 30, '2025-10-28 09:00:00', '2025-10-28 17:00:00');
ERROR 1644 (45000): Error: Only 20 units available for this resource.
```

2. STORED PROCEDURE

Question: Retrieve event summary with venue, organizer, and participant count.

Code:

```
DELIMITER $$
```

```
CREATE PROCEDURE GetEventSummary(IN event_name VARCHAR(255))
BEGIN
SELECT
    e.name AS Event_Name,
    e.date AS Event_Date,
    v.name AS Venue,
    h.name AS Organizer,
    COUNT(p.user_id) AS Total_Participants
    FROM tbl_events e
    JOIN tbl_venues v ON e.location_id = v.id
    JOIN tbl_hosts h ON e.organizer_id = h.id
    LEFT JOIN tbl_event_participants p ON e.id = p.event_id
    WHERE e.name = event_name
    GROUP BY e.id;
END$$
```

```
DELIMITER ;
```

Output Screenshot:

```
mysql> CALL geteventsummary("Hackathon 2025");
```

Event_Name	Event_Date	Venue	Organizer	Total_Participants
Hackathon 2025	2025-10-20	Auditorium	Dr. Ramesh Kumar	1

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)

Question: Automatically updates the availability and quantity of resources based on the current time.

Code:

```
DELIMITER $$
CREATE PROCEDURE ReplenishResources()
BEGIN
    UPDATE tbl_resources r
    JOIN tbl_event_resources er ON r.id = er.resource_id
    SET
        r.is_available = 1,
        r.maintenance_status = 'Available',
        r.quantity = r.quantity + er.quantity_booked
    WHERE er.booking_end < NOW();
    UPDATE tbl_resources r
    JOIN tbl_resource_maintenance m ON r.id = m.resource_id
    SET
        r.is_available = 1,
        r.maintenance_status = 'Available'
    WHERE m.maintenance_end IS NOT NULL AND m.maintenance_end < NOW();
    SELECT
        id AS Resource_ID,
        name AS Resource_Name,
        quantity,
        is_available,
        maintenance_status
    FROM tbl_resources;
END$$

DELIMITER ;
```

Output Screenshot:

mysql> select * from tbl_resources;

id	name	type	quantity	description	is_available	maintenance_status
1	Projector Screen	AV Equipment	10	Portable 100-inch projector screens	1	Available
2	Microphone Set	AV Equipment	20	Wireless microphone with stand	1	Available
3	Tables	Furniture	100	6-foot rectangular tables	1	Available
4	Chairs	Furniture	500	Standard plastic chairs	0	Under Maintenance
5	Sound System	AV Equipment	4	High-power speakers with mixer	1	Booked

5 rows in set (0.00 sec)

mysql> CALL replenishresources();

Resource_ID	Resource_Name	quantity	is_available	maintenance_status
1	Projector Screen	10	1	Available
2	Microphone Set	25	1	Available
3	Tables	150	1	Available
4	Chairs	500	1	Available
5	Sound System	4	1	Available

5 rows in set (0.00 sec)

3. FUNCTION

Question: Calculate average feedback rating for an event.

Code:

```
DELIMITER $$
CREATE FUNCTION GetAverageRating(eventId INT)
RETURNS DECIMAL(3,2)
DETERMINISTIC
BEGIN
    DECLARE avgRating DECIMAL(3,2);
```

```

SELECT AVG(rating) INTO avgRating
FROM tbl_event_feedback
WHERE event_id = eventId;
RETURN IFNULL(avgRating, 0);
END$$
DELIMITER ;

```

Output Screenshot:

```

mysql> select * from tbl_event_feedback;
+-----+-----+-----+-----+-----+-----+
| id | event_id | user_id | rating | comments | submitted_at |
+-----+-----+-----+-----+-----+-----+
| 1 | 4 | 3 | 5 | Excellent workshop, very informative and hands-on. | 2025-10-24 20:40:11 |
| 2 | 1 | 1 | 4 | Well organized, but the internet was a bit slow. | 2025-10-24 20:40:11 |
| 3 | 2 | 2 | 5 | The speaker was amazing! Very inspiring talk. | 2025-10-24 20:40:11 |
| 4 | 4 | 1 | 3 | Could have been better if the duration was longer. | 2025-10-24 20:40:11 |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT GetAverageRating(4) AS 'Average Rating for Event 4 ';
+-----+
| Average Rating for Event 4 |
+-----+
| 4.00 |
+-----+
1 row in set (0.00 sec)

```